

3.18

$$\widehat{INSURANCE}_{(se)} = 6,855 + 3,880 INCOME$$

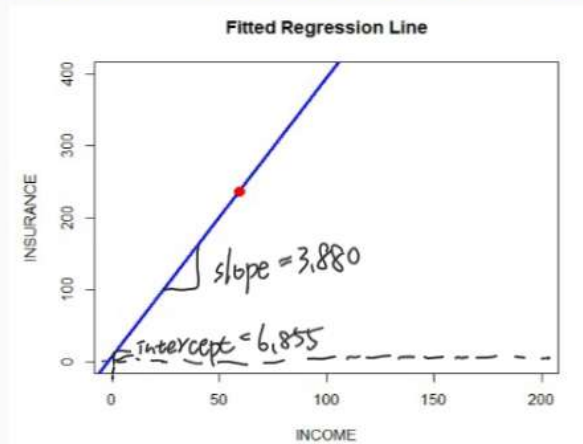
(1,383)      (0,112)

(a)

the estimated slope = 3,880

the estimated intercept = 6,855

the sample mean of the amount of insurance held =  $6,855 + 3,880 \times 59.3 = 236,939$



(b)

$$3,880 \times 1000 = 3880$$

the average amount of insurance held will increase 3880 with each additional 1000 of household income. (point estimate)

$$\alpha = 0.05, df = 18, t(18, 0.975) = 2.101$$

$$\begin{aligned} \text{A 95\% interval estimate is } & [3,880 - 2,101 \times 0,112, 3,880 + 2,101 \times 0,112] \\ & = [3,645, 4,115] \end{aligned}$$

$$\Rightarrow [3,645 \times 1000, 4,115 \times 1000] = [3645, 4115]$$

In repeated sampling, about 95% intervals constructed this way will contain the true value of the increase in insurance held with each additional 1000 of household income.

(c)

$$\widehat{\text{INSURANCE}} = 6.855 + 3.880 \times 100 = 394.855$$

$$\begin{aligned} \text{se}(\widehat{\text{INSURANCE}} | \text{INCOME} = 100) &= \sqrt{\widehat{\text{var}}(b_1) + 100^2 \widehat{\text{var}}(b_2) + 2 \times 100 \times \widehat{\text{cov}}(b_1, b_2)} \\ &= \sqrt{7.383^2 + 10000 \times 0.112^2 + 200 \times (-0.746)} \\ &= \sqrt{30.748689} = 5.5452 \end{aligned}$$

$$\alpha = 0.01, df = 18, t(18, 0.995) = 2.878$$

$$[394.855 - 2.878 \times 5.5452, 394.855 + 2.878 \times 5.5452] = [378.8961, 410.8139]$$

$$\Rightarrow [378.8961 \times 1000, 410.8139 \times 1000] = [378896.1, 410813.9]$$

99% interval estimate of the expected amount of insurance held by a household with 100000 income is  $[378896.1, 410813.9]$

(d)

$$H_0: \beta_2 = 5 \text{ (the statement is true)}$$

$$H_1: \beta_2 \neq 5 \text{ (the statement is not true)}$$

$$\text{The test statistic is } t^* = \frac{b_2 - 5}{\text{se}(b_2)} = \frac{3.88 - 5}{0.112} = -10$$

$$\alpha = 0.05, df = 18, t(18, 0.975) = 2.101$$

$$\text{Rejection region} = \{t: |t| > 2.101\}$$

$$\Rightarrow |-10| = 10 > 2.101, \text{ so we reject } H_0.$$

$\therefore$  We reject  $H_0$  and conclude that the data do not support the claim.



(e)

$$H_0: \beta_2 = 1$$

$$H_1: \beta_2 > 1$$

The test statistic is  $t^* = \frac{b_2 - 1}{se(b_2)} = \frac{3.88 - 1}{0.112} = 25.7143$

$$\alpha = 0.01, df = 18, t(18, 0.99) = 2.552$$

$$\text{Rejection region} = \{t : t > 2.552\}$$

$$\Rightarrow 25.7143 > 2.552, \text{ so we reject } H_0.$$

$\therefore$  We reject  $H_0$  and conclude that as income increases the amount of life insurance held significantly increases by larger amount with  $\beta_2 > 1$ .

**3.23**

(a)

The estimated regression model is  $\widehat{\text{PRICE}} = 93,5659 + 0.1845 \text{ SQFT}^2$   
(se) (6,0722) (0,005256)

$$\text{marginal effect} = \frac{d \text{ PRICE}}{d \text{ SQFT}} = 2 \alpha_2 \text{ SQFT}$$

$$\text{SQFT} = 20 \Rightarrow 2 \times \alpha_2 \times 20 = 40 \alpha_2$$

$$H_0: 40 \alpha_2 \leq 13 \quad (\alpha_2 \leq 0.325)$$

$$H_1: 40 \alpha_2 > 13 \quad (\alpha_2 > 0.325)$$

The test statistic is  $t^* = \frac{\hat{\alpha}_2 - 0.325}{se(\hat{\alpha}_2)} = \frac{0.1845 - 0.325}{0.005256} = -26.73$

$$\alpha = 0.05, df = 498, t(498, 0.95) = 1.645$$

$$\text{Rejection region} = \{t : t > 1.645\}$$

$$\Rightarrow -26.73 < 1.645, \text{ so we do not reject } H_0.$$

$$p\text{-value} = P(t_{(498)} > -26.73) \cong 1$$

$$\Rightarrow p\text{-value} \cong 1 > \alpha = 0.05, \text{ so we do not reject } H_0.$$

$\therefore$  We can not conclude that the marginal effect is significantly greater than 13000.

(b)

$$\text{marginal effect} = \frac{d\text{PRICE}}{d\text{SQFT}} = 2\alpha_2 \text{SQFT}$$

$$\text{SQFT} = 40 \Rightarrow 2 \times \alpha_2 \times 40 = 80\alpha_2$$

$$H_0: 80\alpha_2 \leq 13 \quad (\alpha_2 \leq 0.1625)$$

$$H_1: 80\alpha_2 > 13 \quad (\alpha_2 > 0.1625)$$

$$\text{The test statistic is } t^* = \frac{\hat{\alpha}_2 - 0.1625}{\text{se}(\hat{\alpha}_2)} = \frac{0.1845 - 0.1625}{0.005256} = 4.19$$

$$\alpha = 0.05, df = 498, t(498, 0.95) = 1.645$$

$$\text{Rejection region: } \{t: t > 1.645\}$$

$$\Rightarrow 4.19 > 1.645, \text{ so we reject } H_0.$$

$$p\text{-value} = P(t_{(498)} > 4.19) = 0.00001654$$

$$\Rightarrow p\text{-value} = 0.00001654 < \alpha = 0.05, \text{ so we reject } H_0.$$

$\therefore$  We conclude that the marginal effect is significantly greater than 13000.

(c)

$$\hat{E}(\text{PRICE} | \text{SQFT} = 20) = 93.5659 + 0.1845 \times 20^2 = 167.3735 \text{ (thousands)}$$

$$\begin{aligned} \text{se}(\hat{\text{PRICE}} | \text{SQFT} = 20) &= \sqrt{\text{var}(\hat{\alpha}_1) + 20^4 \cdot \text{var}(\hat{\alpha}_2) + 2 \cdot 20^2 \cdot \text{cov}(\hat{\alpha}_1, \hat{\alpha}_2)} \\ &= 4.746378 \end{aligned}$$

$$t(498, 0.975) = 1.964739$$

$$\begin{aligned} &[167.3735 - 1.964739 \times 4.746378, 167.3735 + 1.964739 \times 4.746378] \\ &= [158.0481, 176.6988] \text{ (thousands)} \end{aligned}$$

$$\Rightarrow \text{A 95\% interval estimate of the expected price is } [158048.1, 176698.8]$$

In repeated sampling, about 95% intervals constructed this way will contain the true value of the price for a house of 2000 square feet.



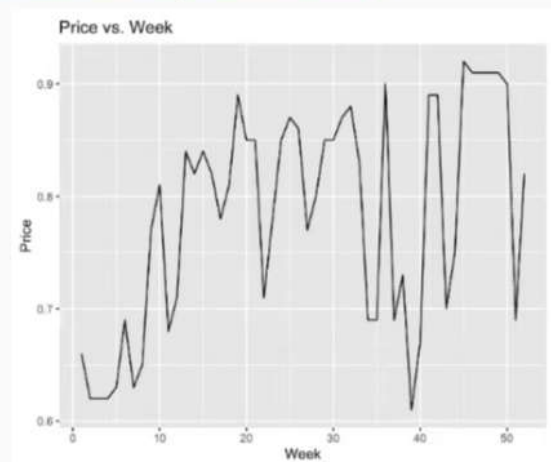
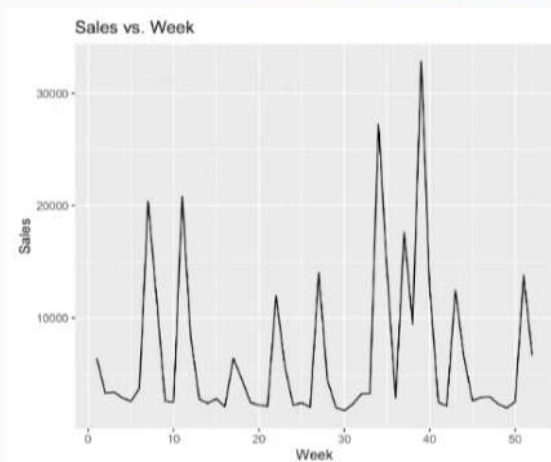
(d)

There are three houses with 2000 square feet of living area in the sample. Their selling prices are 138000, 169000 and 183000. The sample mean of their selling prices is 163333.33. It is compatible with the interval estimate  $[158048.1, 176698.8]$  in part (c).

3.31

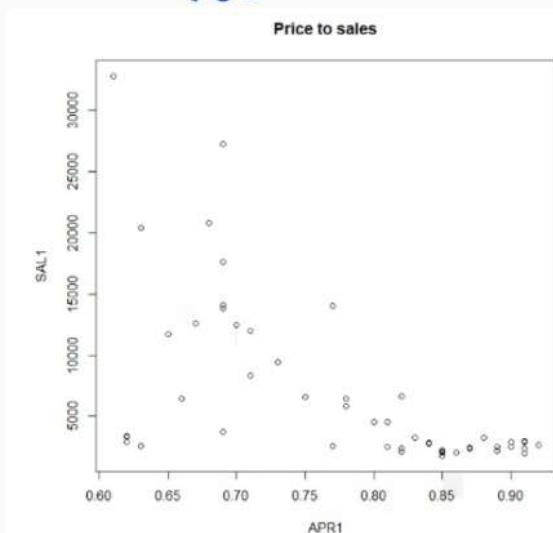
(a)

| variable | Mean    | Min  | Max   | SD      |
|----------|---------|------|-------|---------|
| SAL1     | 6718.71 | 1762 | 32820 | 6915.08 |
| APR1     | 0.78    | 0.61 | 0.92  | 0.1     |



The variations are highly correlated. The sharp drops in price correspond almost perfectly with the massive spikes in sales. This indicates that the high week-to-week variation is likely driven by promotional discounts.

(b)



There is an inverse relationship between the weekly sales and weekly price.

It is what we expected because of the law of demand. (When price  $\uparrow$ , the demand  $\downarrow$  and the sales  $\downarrow$ . When price  $\downarrow$ , the demand  $\uparrow$  and the sales  $\uparrow$ .)

(c)

The estimated regression model is  $\widehat{SAL} = 40714.22 - 434.45 \text{ PRICE}$   
(se) (6196.42) (78.58)

The effect of a one cent increase in the price means that the marginal effect of PRICE, which is  $\frac{\partial \widehat{SAL}}{\partial \text{PRICE}} = \beta_2$ . The point estimator for  $\beta_2$  is  $b_2 = -434.45$ .

$$\alpha = 0.05, df = 52 - 2 = 50, t(50, 0.975) = 2.009$$
$$[-434.45 - 2.009 \times 78.58, -434.45 + 2.009 \times 78.58]$$
$$= [-592.2897, -276.6049]$$

$\therefore$  A 95% interval estimate is  $[-592.2897, -276.6049]$

(d)

$$70 \text{ cents} \Rightarrow \text{PRICE} = 70$$

$$\Rightarrow \widehat{SAL} = 40714.22 - 434.45 \times 70 = 10302.9$$

$$se(\widehat{Y}_i | X=70) = \sqrt{MSE \left( \frac{1}{n} + \frac{(70 - \bar{X})^2}{S_{XX}} \right)} = 1001.231$$

$$\alpha = 0.1, df = 50, t(50, 0.95) = 1.676$$

$$[10302.9 - 1.676 \times 1001.231, 10302.9 + 1.676 \times 1001.231]$$
$$= [8624.934, 11980.87]$$

$\therefore$  A 90% interval estimate is  $[8624.934, 11980.87]$



(e)

$$\widehat{SAL} = 40714.22 - 434.45 \text{ PRICE} \quad , \text{ mean}(\text{PRICE}) = 78.25,$$

(se) (6196.42) (78.58)

$$\text{mean}(\widehat{SAL}) = 6718.712, \quad \hat{\epsilon} = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = m \cdot \frac{x}{y} = \beta_2 \cdot \frac{\text{PRICE}}{\widehat{SAL}} = (-434.45) \cdot \frac{78.25}{6718.712} = -5.0598$$

$$\text{se}(\hat{\epsilon}) = \text{se}(\beta_2) \times \frac{\text{PRICE}}{\widehat{SAL}} = 78.58 \times \frac{78.25}{6718.712} = 0.9152$$

$$\alpha = 0.05, \quad df = 50, \quad t(50, 0.975) = 2.009$$

$$[-5.0598 - 2.009 \times 0.9152, -5.0598 + 2.009 \times 0.9152] = [-6.8981, -3.2215]$$

$\therefore$  A 95% interval estimate is  $[-6.8981, -3.2215]$

$\hat{\epsilon} = -5.0598 \Rightarrow$  The sales are highly elastic. This makes economic sense because canned tuna has many substitutes in the market. Consumers have many alternatives to choose from if the price of canned tuna increases. Therefore, the degree of elasticity is high.

(f)

$$H_0: \epsilon = -3$$

$$H_1: \epsilon \neq -3$$

$$\text{The test statistic is } t^* = \frac{\hat{\epsilon} - (-3)}{\text{se}(\hat{\epsilon})} = \frac{-2.0598}{0.9152} = -2.25$$

$$\alpha = 0.1, \quad df = 50, \quad t(50, 0.95) = 1.676$$

$$\text{Rejection region} = \{t: |t| > 1.676\}$$

$$\Rightarrow |-2.25| = 2.25 > 1.676, \text{ so we reject } H_0.$$

$$p\text{-value} = P(t_{(50)} < -2.25) \times 2 = 0.0288$$

$$\Rightarrow p\text{-value} = 0.0288 < \alpha = 0.1, \text{ so we reject } H_0.$$

$\therefore$  We conclude that the elasticity is significantly different from  $-3$ .